

Concentration lower-bound score, stated first. At window w , score a candidate right direction $v \in \mathbb{S}^{d-1}$ by

$$\text{score}_w^{\text{conc}}(v) := \frac{E_w(v)}{p_w} \left(1 - \frac{c_\delta}{\sqrt{k_{\text{eff}}^{(R,w)}(v)}} \right)_+, \quad (1)$$

where

$$E_w(v) := \|Bv\|_2^2 + \|A_w v\|_2^2, \quad (2)$$

$$k_{\text{eff}}^{(R,w)}(v) := \frac{\|C_w v\|_2^4}{\|C_w v\|_4^4} = e^{h_2^{(R,w)}(v)}, \quad (3)$$

$$C_w := \begin{pmatrix} R \\ A_w \end{pmatrix} \in \mathbb{R}^{(s+k) \times d}, \quad p_w := \frac{n_{\text{seen}} + k}{n}, \quad (4)$$

and $c_\delta > 0$ is a confidence multiplier controlling the failure probability. The operator $(x)_+ := \max\{x, 0\}$ clamps the score at zero when the bound is vacuous. If this is the first window and no old row sample is available, take $R = \emptyset$ and $s = 0$; if no old sketch is available, take $B = 0$ and $n_{\text{seen}} = 0$.

Notation in detail. Let $A \in \mathbb{R}^{n \times d}$ be the full data matrix, let A_w denote the current row-window with k rows, let $B \in \mathbb{R}^{r \times d}$ denote the sketch carried from previous windows, and let $R \in \mathbb{R}^{s \times d}$ denote a raw row memory. Write $v \in \mathbb{S}^{d-1}$, $x := Av \in \mathbb{R}^n$, $y := A_w v \in \mathbb{R}^k$, and $T := \|x\|_2^2 = \|Av\|_2^2$. Denote the number of raw rows represented after the current window and the seen-fraction by

$$m_w := n_{\text{seen}} + k, \quad p_w := \frac{m_w}{n}. \quad (5)$$

Assume $s + k \geq 2$ so that the pooled participation ratio is well-defined and $p_w \in (0, 1]$.

Motivation: bound and argmax, simultaneously. The combined lower-bound score $E_w(v) \cdot p_w^{-h_2^{(R,w)}(v)/(s+k)}$ uses entropy as a multiplicative reward, with an exponent $1/(s+k)$ whose argmax is empirically dominated by the E_w factor on structured matrices. The subset-mass upper-bound score $E_w(v)/(1 - \psi_w(v))$ is monotone in the wrong direction: its argmax prefers low entropy. Neither is simultaneously a tight numerical estimator of $\|Av\|_2^2$ and a selection criterion whose argmax approaches the oracle direction.

This note derives an objective that is both a valid (high-probability) numerical lower bound on $\|Av\|_2^2$ and a selection criterion whose argmax favors flat directions where the bound is informative, from the standard unbiased plug-in estimator and a one-sided Chebyshev inequality.

Setup: unbiased block estimator. Specialize first to the one-block, no-memory case ($B = 0$, $R = \emptyset$, $s = 0$, $n_{\text{seen}} = 0$) for clarity; the pooled form

follows at the end. Draw the k window rows i.i.d. uniformly from the n full rows (with replacement; the with-/without-replacement distinction is lower order for $k \ll n$). For any row i ,

$$\Pr[i \in B] = \frac{k}{n} = p_w. \quad (6)$$

Hence

$$\mathbb{E}[\|A_w v\|_2^2] = \sum_{i=1}^n x_i^2 \cdot \Pr[i \in B] = p_w T, \quad (7)$$

so the rescaled block estimator

$$S := \frac{\|A_w v\|_2^2}{p_w} \quad (8)$$

satisfies $\mathbb{E}[S] = T$.

Variance in closed form. Write $\|A_w v\|_2^2 = \sum_{j=1}^k \xi_j$ with $\xi_j = x_{I_j}^2$ for i.i.d. uniform row indices $I_j \in [n]$. Each ξ_j has $\mathbb{E}[\xi_j] = T/n$ and

$$\text{Var}(\xi_j) = \frac{1}{n} \sum_i x_i^4 - \left(\frac{T}{n}\right)^2 = \frac{\|x\|_4^4}{n} - \frac{T^2}{n^2}. \quad (9)$$

Therefore

$$\text{Var}(\|A_w v\|_2^2) = k \cdot \text{Var}(\xi_j) = p_w \left(\|x\|_4^4 - \frac{T^2}{n} \right), \quad (10)$$

and

$$\text{Var}(S) = \frac{1}{p_w^2} \text{Var}(\|A_w v\|_2^2) = \frac{1}{p_w} \left(\|x\|_4^4 - \frac{T^2}{n} \right). \quad (11)$$

Using the full-matrix participation ratio $k_{\text{eff}}(x) := T^2/\|x\|_4^4$,

$$\text{Var}(S) = \frac{T^2}{p_w} \left(\frac{1}{k_{\text{eff}}(x)} - \frac{1}{n} \right) = T^2 \cdot \frac{n - k_{\text{eff}}(x)}{k \cdot k_{\text{eff}}(x)}. \quad (12)$$

Sanity checks:

- Flat direction, $k_{\text{eff}}(x) = n$: $\text{Var}(S) = 0$. The estimator concentrates perfectly, since every row contributes identically.
- Spike direction, $k_{\text{eff}}(x) = 1$: $\text{Var}(S) = T^2(n-1)/k \approx T^2/p_w$. The standard deviation is on the order of T itself, so single-block estimation is uninformative.

One-sided Chebyshev lower bound. By Chebyshev's inequality,

$$\Pr[T \geq S + t] \leq \Pr[|S - T| \geq t] \leq \frac{\text{Var}(S)}{t^2}. \quad (13)$$

Setting the right-hand side equal to δ and solving, $t = \text{SD}(S)/\sqrt{\delta}$. Rearranging the event,

$$\Pr\left[T \geq S - \frac{\text{SD}(S)}{\sqrt{\delta}}\right] \geq 1 - \delta. \quad (14)$$

Wait—this is the wrong side; for a lower bound on T we want the upper tail $\Pr[S - T \leq -t]$. By symmetry of Chebyshev in $|S - T|$, the same inequality applies:

$$\Pr\left[T \geq S \cdot \left(1 - \frac{\text{SD}(S)/S}{\sqrt{\delta}}\right)\right] \geq 1 - \delta, \quad (15)$$

provided the right-hand side is treated as a lower estimate rather than a numerical evaluation of T . Using the variance formula above,

$$\frac{\text{SD}(S)}{T} = \sqrt{\frac{n - k_{\text{eff}}(x)}{k \cdot k_{\text{eff}}(x)}}. \quad (16)$$

Plug-in estimator. Both T and $k_{\text{eff}}(x)$ are unobservable. Replace T by S (making the confidence width self-referential) and replace $k_{\text{eff}}(x)$ by the block participation ratio $k_{\text{eff}}(y) := \|y\|_2^4 / \|y\|_4^4$ where $y = A_w v$. Under the dimension-normalized entropy heuristic $h_2(x)/\log n \approx h_2(y)/\log k$, these plug-ins agree to leading order for flat directions and both collapse to 1 for concentrated spike directions, so the flat/spike scaling of the confidence width is preserved.

For $k_{\text{eff}}(y) \ll n$, drop the $n - k_{\text{eff}}$ correction in the numerator to get the compact form

$$\frac{\text{SD}(S)}{S} \approx \sqrt{\frac{1}{p_w k_{\text{eff}}(y)}}, \quad (17)$$

and the Chebyshev LB becomes

$$T \gtrsim S \cdot \left(1 - \frac{c_\delta}{\sqrt{k_{\text{eff}}(y)}}\right), \quad c_\delta = \frac{1}{\sqrt{p_w \delta}}. \quad (18)$$

Clamping at zero to avoid reporting negative lower bounds yields the one-block score

$$\text{score}_1^{\text{conc}}(v) = \frac{\|A_w v\|_2^2}{p_w} \left(1 - \frac{c_\delta}{\sqrt{k_{\text{eff}}(y)}}\right)_+. \quad (19)$$

Interpretation and argmax behavior. The score has three regimes determined entirely by $k_{\text{eff}}(y)$:

- **Flat direction** ($k_{\text{eff}}(y) \approx k$): $c_\delta/\sqrt{k}$ is small, so the multiplicative correction is near 1 and $\text{score}^{\text{conc}} \approx \|A_w v\|_2^2/p_w \approx T$. The bound is tight and the value agrees with the naive unbiased estimator.
- **Moderately concentrated** ($c_\delta^2 < k_{\text{eff}}(y) < k$): the correction discounts the estimate but remains positive. The bound is informative, with width proportional to $1/\sqrt{k_{\text{eff}}}$.
- **Spike direction** ($k_{\text{eff}}(y) \leq c_\delta^2$): $c_\delta/\sqrt{k_{\text{eff}}} \geq 1$, the bracket clamps to 0, and the direction is declared *vacuous*. The single-block data does not support any positive lower bound on T ; the score refuses to commit.

The argmax of $\text{score}^{\text{conc}}$ therefore selects directions on which the block evidence is both high-energy *and* sufficiently spread. Spike-like directions are filtered out not by an adversarial penalty but by the statistical impossibility of certifying their energy from one block.

Fundamental obstruction to single-block tight bounds on spikes. The variance formula $\text{Var}(S) = T^2(n - k_{\text{eff}}(x))/(k \cdot k_{\text{eff}}(x))$ shows that the standard deviation of the unbiased estimator S is $\Omega(T)$ whenever $k_{\text{eff}}(x) = O(1)$. Any estimator formed from one block of size $k < n$ inherits this variance floor, so no single-block two-sided bound can be tight on a pure spike direction. The concentration score captures this obstruction by design: it is tight where Chebyshev is informative and explicitly vacuous where it is not.

Tighter confidence constants via Bernstein. The Chebyshev constant $c_\delta = (p_w \delta)^{-1/2}$ scales polynomially in $1/\delta$. For small target failure probabilities, Bernstein’s inequality for bounded i.i.d. summands tightens this to

$$c_\delta^{\text{Bern}} = \sqrt{2 \log(1/\delta) \cdot \frac{1 - p_w}{p_w}} + \frac{\log(1/\delta)}{3} \cdot \frac{\max_i x_i^2/T}{\sqrt{k_{\text{eff}}(y)}}, \quad (20)$$

scaling as $O(\sqrt{\log(1/\delta)})$ up to the leverage term $\max_i x_i^2/T$. The leverage term can be bounded in practice by the block-observed leverage $\max_j y_j^2/\|y\|_2^2$ when the block is large enough to contain the dominant row with high probability; otherwise the Chebyshev form is preferred because Bernstein becomes dominated by the leverage term.

Pooled streaming form with sketch and row memory. Restoring the general window w , replace the single-block energy by the seen energy surrogate $E_w(v) = \|Bv\|_2^2 + \|A_w v\|_2^2$ and the block participation ratio by the pooled one,

$$k_{\text{eff}}^{(R,w)}(v) := \frac{\|C_w v\|_2^4}{\|C_w v\|_4^4}, \quad C_w = \begin{pmatrix} R \\ A_w \end{pmatrix}. \quad (21)$$

The unbiased estimator becomes $E_w(v)/p_w$; its variance is reduced relative to one block because R supplies additional independent row samples. Substituting into the Chebyshev LB and plugging in the pooled k_{eff} gives

$$\text{score}_w^{\text{conc}}(v) = \frac{E_w(v)}{p_w} \left(1 - \frac{c_\delta}{\sqrt{k_{\text{eff}}^{(R,w)}(v)}} \right)_+. \quad (22)$$

Exact gradient. Let $y_B := Bv$, $z := A_w v$, $y := C_w v$, and assume $E_w(v) > 0$, $\|y\|_2 > 0$, $\|y\|_4 > 0$, and $c_\delta < \sqrt{k_{\text{eff}}^{(R,w)}(v)}$ (bracket strictly positive). Write

$$L(v) := \left(1 - \frac{c_\delta}{\sqrt{k_{\text{eff}}^{(R,w)}(v)}} \right). \quad (23)$$

The energy gradient is $\nabla E_w(v) = 2B^\top Bv + 2A_w^\top A_w v$. The pooled entropy gradient is

$$\nabla h_2^{(R,w)}(v) = 4 \frac{C_w^\top C_w v}{\|C_w v\|_2^2} - 4 \frac{C_w^\top (C_w v)^{\odot 3}}{\|C_w v\|_4^4}, \quad (24)$$

and since $k_{\text{eff}}^{(R,w)}(v) = e^{h_2^{(R,w)}(v)}$,

$$\nabla \log k_{\text{eff}}^{(R,w)}(v) = \nabla h_2^{(R,w)}(v). \quad (25)$$

Differentiating L ,

$$\nabla L(v) = \frac{c_\delta}{2 \sqrt{k_{\text{eff}}^{(R,w)}(v)}} \nabla h_2^{(R,w)}(v). \quad (26)$$

Taking logs of the (positive-bracket) score, $\log \text{score}_w^{\text{conc}}(v) = \log E_w(v) - \log p_w + \log L(v)$, and differentiating,

$$\nabla \log \text{score}_w^{\text{conc}}(v) = \frac{\nabla E_w(v)}{E_w(v)} + \frac{1}{L(v)} \frac{c_\delta}{2 \sqrt{k_{\text{eff}}^{(R,w)}(v)}} \nabla h_2^{(R,w)}(v). \quad (27)$$

Equivalently,

$$\nabla \text{score}_w^{\text{conc}}(v) = \text{score}_w^{\text{conc}}(v) \left[\frac{2B^\top Bv + 2A_w^\top A_w v}{E_w(v)} + 2\lambda_w(v) \left(\frac{C_w^\top C_w v}{\|C_w v\|_2^2} - \frac{C_w^\top (C_w v)^{\odot 3}}{\|C_w v\|_4^4} \right) \right], \quad (28)$$

with direction-dependent coupling

$$\lambda_w(v) := \frac{c_\delta}{L(v) \sqrt{k_{\text{eff}}^{(R,w)}(v)}} = \frac{c_\delta}{\sqrt{k_{\text{eff}}^{(R,w)}(v)} - c_\delta}. \quad (29)$$

The sign of the entropy contribution coincides with $\nabla h_2^{(R,w)}(v)$, so the gradient pushes v toward *higher* pooled entropy. $\lambda_w(v)$ is strictly positive in the feasible region $k_{\text{eff}}^{(R,w)}(v) > c_\delta^2$ and diverges as the direction approaches the vacuity boundary, so the entropy reward intensifies as the confidence width approaches its maximum.

Expanded row-memory form. Because $C_w^\top C_w v = R^\top R v + A_w^\top A_w v$ and $C_w^\top (C_w v)^{\odot 3} = R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}$, the pooled gradient can be evaluated without forming C_w :

$$\frac{C_w^\top C_w v}{\|C_w v\|_2^2} = \frac{R^\top R v + A_w^\top A_w v}{\|R v\|_2^2 + \|A_w v\|_2^2}, \quad (30)$$

$$\frac{C_w^\top (C_w v)^{\odot 3}}{\|C_w v\|_4^4} = \frac{R^\top (R v)^{\odot 3} + A_w^\top (A_w v)^{\odot 3}}{\|R v\|_4^4 + \|A_w v\|_4^4}. \quad (31)$$

Optimization problem and feasibility. Extract directions sequentially by solving

$$\max_{v \in \mathbb{S}^{d-1}, Q^\top v=0, k_{\text{eff}}^{(R,w)}(v) > c_\delta^2} \text{score}_w^{\text{conc}}(v), \quad (32)$$

where the columns of Q are previously accepted directions. The feasibility constraint $k_{\text{eff}}^{(R,w)}(v) > c_\delta^2$ is equivalent to $h_2^{(R,w)}(v) > 2 \log c_\delta$ and has a meaningful minimum at the confidence threshold. For $n = 128$, $k = 32$, $s = 0$, and $\delta = 0.1$: $c_\delta = (p_w \delta)^{-1/2} = (0.25 \cdot 0.1)^{-1/2} = \sqrt{40} \approx 6.32$, so $c_\delta^2 \approx 40$, requiring $k_{\text{eff}}^{(R,w)}(v) \gtrsim 40$. Since the first-window pooled pool has only $k = 32$ rows, the constraint is unsatisfiable under Chebyshev at $\delta = 0.1$ on this problem size, and a Bernstein constant must be used or δ relaxed.

Projected full-gradient ascent. Within the feasible set, use the exact gradient projected onto the tangent space of the sphere and the column complement of Q :

$$g_{\text{tan}}(v) = (I - Q Q^\top - v v^\top) \nabla \text{score}_w^{\text{conc}}(v). \quad (33)$$

Backtracking Armijo on $\text{score}^{\text{conc}}$ yields a monotone-ascent solver. Reject any step that crosses the feasibility boundary $k_{\text{eff}}^{(R,w)}(v) = c_\delta^2$: the step either lands in the clamped region (where the score is constantly zero) or runs into the non-differentiable join.

Frame-level variant. For rank- r extraction, the frame estimator $S_{\text{frame}} := \sum_{j=1}^r \|A_w v_j\|_2^2 / p_w$ has variance

$$\text{Var}(S_{\text{frame}}) = \frac{1}{p_w^2} k \text{Var}_i \left(\sum_{j=1}^r (A v_j)_i^2 \right), \quad (34)$$

which involves the frame row-response $\sum_j x_{i,j}^2$ instead of the single-direction x_i^2 . When the retained directions share spike rows, the inter-direction correlations reduce the frame variance below the sum of per-direction variances, so a frame-level concentration LB can be tighter than $\sum_j \text{score}^{\text{conc}}(v_j)$. This yields a coupled Stiefel objective

$$\text{score}_w^{\text{frame,conc}}(V) = \frac{\sum_j E_w(v_j)}{p_w} \left(1 - \frac{c_\delta}{\sqrt{k_{\text{eff}}^{(R,w,r)}(V)}} \right)_+, \quad (35)$$

with $V = [v_1, \dots, v_r]$ on the Stiefel manifold and frame-level participation ratio

$$k_{\text{eff}}^{(R,w,r)}(V) := \frac{\left(\sum_i (\sum_j (C_w V)_{i,j}^2) \right)^2}{\sum_i (\sum_j (C_w V)_{i,j}^2)^2}. \quad (36)$$

The derivation is unchanged: Chebyshev on the frame-level unbiased estimator, variance via the frame row-response, plug-in using pooled quantities.

Practical remarks. The confidence constant c_δ must be chosen so that the feasible region contains plausibly-informative directions; with Chebyshev on small pools this can be restrictive, and Bernstein or Hoeffding forms are preferable when the block-observed leverage $\max_j y_j^2 / \|y\|_2^2$ is small. The clamp $(\cdot)_+$ should be treated as a soft gate in continuous optimization: replace it by the smoothed alternative $(\cdot) \cdot \sigma(\alpha \cdot \cdot)$ with σ a sigmoid and α large, or carry the pre-clamp score plus a separate feasibility flag.

The concentration score is not a replacement for the combined lower-bound score; it is a complement. The combined score extracts per-direction information from the full window participation-ratio reward; the concentration score refuses to commit to directions where the single-window evidence is statistically insufficient. A natural two-stage pipeline is therefore: (i) use $\text{score}^{\text{conc}}$ to lock flat directions where the bound is tight, then (ii) use $\text{score}^{\text{comb}}$ (or its oracle-informed variants) on the residual operator to hunt spike directions that require pooled multi-window evidence.